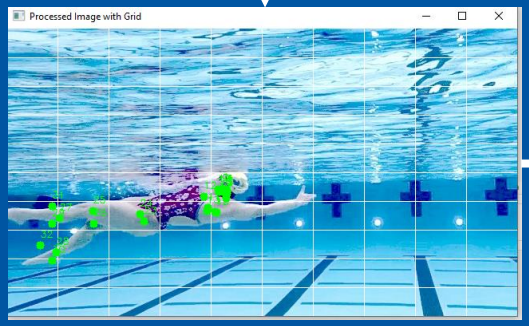
General Overview of Image Processing Pipeline

1. Extract Images

2. Enhance Image



3. MediaPipe Object Location

4. SAM Segmentation

5. Overlay Segmented Swimmer

6. Run Pose Estimation Using YOLOv7

- Numbers 1 – 6 display the general pipeline of image processing used to execute the task.
- Each image is extracted and enhance as preprocessing to be sent MediaPipe neural network.
- The main use of using MediaPipe is to locate where in the enhanced image the swimmer might be in frame.
- The YOLOv7 provided critical key-points data found for the segmented swimmer to be used for training.